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ASSIGNMENT 2

PROGRESS 2 – STATE SPACE SEARCH

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INTRODUCTION

The AI Task Automation System is designed to enhance industrial efficiency by automating routine tasks and intelligently managing workflows. Many industries still depend on manual processes, which lead to slow task completion, human errors, and weak coordination across departments. To address these challenges, the system uses Artificial Intelligence to analyze operational conditions and decide the most suitable actions at any moment.

State Space Search plays a central role in this decision-making process. By modeling industrial operations as a collection of states—such as task assignments, machine conditions, resource availability, workflow progress, and anomaly detection—the AI can explore possible transitions and select the most effective path toward achieving the goal state. This structured approach allows the system to navigate complex, changing environments where tasks, resources, and machine performance constantly shift.

Incorporating State Space Search enables the AI to identify delays, resolve bottlenecks, automate workflows, and react quickly to unexpected events. Each previously defined Knowledge Representation (KR)—including Task Assignment, Predictive Maintenance, Resource Optimization, Workflow Automation, and Anomaly Detection—supports and shapes the state transitions, rules, and decision criteria.

By integrating these KR elements with state space search, the AI Task Automation System can determine optimal actions, minimize human error, maintain machine reliability, and ensure smooth, productive industrial operations. This makes the overall system more adaptive, efficient, and aligned with the goals of smart industry transformation.

OVERVIEW OF STATES AND ACTIONS IN SPACE SEARCH

The system is designed to optimize industrial operations by automating repetitive tasks, improving coordination, reducing errors, and enabling predictive maintenance. To achieve this, we model the industrial environment as a state space, where each state represents a snapshot of the system, and actions define the transitions between these states.

States of the System

A state represents the current condition or position of a task at a particular moment within the system. Each state indicates what phase the task is in whether it has been created, assigned, is in progress, or completed. By defining states clearly, the system is able to monitor task progression accurately and identify the necessary steps required to move forward in the workflow.

The following states represent the different stages a task undergoes in the AI Task Automation System:

State	State Name	Description	Attributes	Example
S0	Task	Represents the current status and characteristics of each task in the system. Tasks include production operations, document approvals, inventory updates, and maintenance scheduling (supports KR1 and KR4).	- TaskID - TaskStatus - AssignedTo - Priority - DependsOn	Task 101: Check conveyor condition Status: Pending AssignedTo: Woker 5 Priority: High DependsOn: Task 100
S1	Availability	Represents which employee/machine are currently available to perform tasks. It ensures efficient allocation of personnel and equipment (supports KR1 and KR3).	- Employee/MachineID - Availability - SkillLevel - CurrentTask	Machine 2: Convoyer Availability: Available Skill: Assembly Task: None Worker 5: Sarah Smith Availability: Busy Skill: Machine Operation Task: Task 101
S2	Machine Condition	Monitors the physical condition and operational health of machines to enable predictive maintenance (supports KR2).	- MachineID - Temperature - Vibration - MaintenanceRequired - LastUpdateDate	Machine 2: “Convoyer” Temperature: 85°C Vibration: 12 units MaintenanceRequired: Yes LastUpdateDate: 01/11/25
S3	Resource Utilization	Represents the consumption levels of resources such as materials, energy, and workforce hours (supports KR3).	- ResourceID - UsageLevel - OptiomalLevel - AlertSent	Resource 4: Iron UsageLevel: 92% OptiomalLevel: 99% Alert: False
S4	Workflow	Represents the progress of interdependent tasks across departments or production lines (supports KR4).	- TaskID - TaskStatus - NextTaskReady	Task 100: Prepare Iron Status: Completed NextTask: Task 101
S5	Alert	Captures unusual or abnormal events in operations that could disrupt efficiency (supports KR5).	- ProcessID - Condition - AlertSent - AlertRecipient	Process 2: Task 100 →Task 101 Condition: Normal Alert: False AlertRecipient: Worker 5
S6	Goal State	Represents the final state of the system, where all tasks are completed, machine in noremal condition, and no alert triggered	- TaskID - MachineID - MaintenanceRequired - AlertSent	Task 101: Check conveyor condition Machine 2: “Convoyer” MaintenanceRequired: No Alert: False

Actions of the System

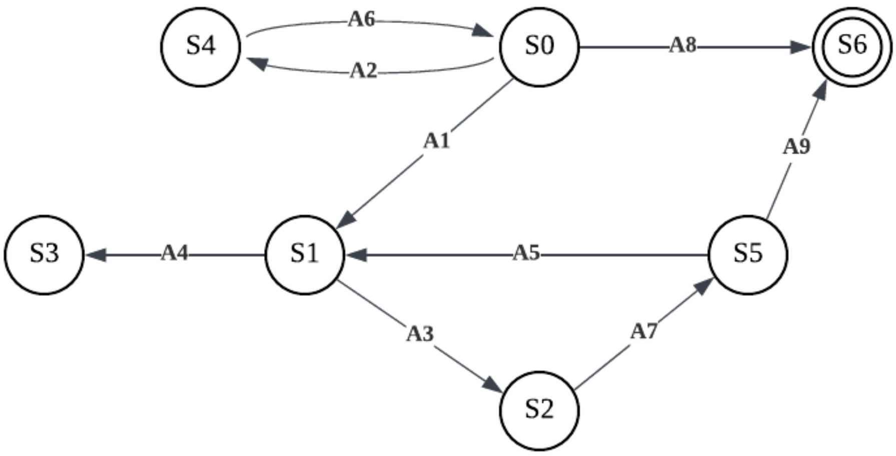
Actions define transitions between states. They are the operations the AI system can perform to move the system closer to its goals. Without actions, states would remain unchanged, and the task would not advance toward completion.

Each action is associated with one or more Knowledge Representations (KR) defined earlier. The actions defined below outline the automated operations that enable movement between states within the system:

Action	Action Name	Precondition	Effect
A1	Assign Task to Worker/Machine (KR1)	- Task is pending - Worker/Machine is available - Skill matches task requirements	- Updates TaskState → In Progress - Assigns worker/machine - Updates Worker/Machine availability
A2	Trigger Dependent Workflow (KR4)	- Preceding task is completed	- Automatically starts dependent tasks - Updates WorkflowState - Ensures smooth operation across departments
A3	Monitor Machine Condition (KR2)	- Machine temperature > threshold OR vibration > limit	- Updates Machine ConditionState - Suggests preventive action
A4	Optimize Resource Usage (KR3)	- Resource usage > optimal level	- Generates recommendations to reduce consumption - Adjusts workflow or task allocation if necessary
A5	Detect and Report Anomalies (KR5)	- Process deviates from normal operation	- Sends notification to supervisors - Updates AnomalyState - Suggests corrective actions
A6	Dependent Workflow (KR4)	- Track the next task in progress	- TaskState → Completed - Triggers dependent tasks - Updates resource and workflow states
A7	Alert Anomalies (KR2)	- Machine temperature > threshold OR vibration > limit - Resource usage > optimal level	- Sends alert to maintenance team
A8	Task Complement (KR4)	- No more tasks left in process	- The system has finished all tasks
A9	Alert No Anomalies (KR2)	- Machine temperature < threshold OR vibration < limit - Resource usage < optimal level	- No anomalies are detected - No alerts are sent

GRAPH OF STATE SPACE

Each node represents a system state, and each directed edge represents an action that moves the system from one state to another. Below is the representation of the system state space in graph:



<Figure 1: State Space Graph>

PROBLEM

In our AI Task Automation System, the issue can be framed as a state space search, where the system moves from a less effective, manual industrial process (initial state) to a completely automated, enhanced, and AI-powered process (goal state).

Initial State:

The starting state reflects the current industrial environment, which is largely reliant on manual labor and is detailed in the proposal's Empathize and Define sections.

Initial State (S_0) includes:

- **Pending tasks:** All tasks (manual operations, document approvals, maintenance, reporting) are unassigned and waiting to be executed.
- **Available workers and machines:** Each worker or machine has its current status (available or busy).
- **Machine and equipment conditions:** Normal operational status without failures, though some machines may require monitoring.
- **Resource levels:** Materials, energy, and inventory at current usage levels.
- **Workflow status:** No dependent tasks have started yet.

Actions:

Actions are the exact actions that an AI system can do to go from one state to another.

A = {A1, A2, A3, A4, A5, A6, A7, A8, A9}

1. **Assign task to worker/machine:** Allocate a pending task based on priority, skill match, and availability (supports KR1).
2. **Trigger dependent tasks:** Start a task automatically when its prerequisites are completed (supports KR4).
3. **Monitor machine and schedule maintenance:** Evaluate sensor data and trigger predictive maintenance if thresholds are exceeded (supports KR2).
4. **Optimize resource usage:** Adjust energy, material, or manpower allocation to stay within optimal limits (supports KR3).
5. **Detect anomalies:** Identify abnormal process behavior and notify supervisors (supports KR5).
6. **Dependent Workflow:** Allocate the next step of an in-progress workflow. After particular task completed perform next depended task (support KR4).
7. **Alert Anomalies:** Sends notifications to supervisors or maintenance teams when a critical anomaly is detected (machine or process-related) (support KR2).
8. **Task Complement:** Finalizes a task after verification of correct execution, and check that all tasks are completed (support KR4),
9. **Alert No Anomalies:** Confirms system stability after monitoring and clears previous warnings if conditions return to normal (support KR2).

Goals:

The goal state explains the ideal completely automated and intelligent system, which aligns with the proposal's Goals of AI Solution section.

- **All tasks completed:** Every production, administrative, and maintenance task is finished.
- **Machines in optimal condition:** No breakdowns or errors, with predictive maintenance completed if required.
- **Resources efficiently used:** Energy, materials, and workforce are allocated optimally, avoiding wastage.
- **No anomalies pending:** All abnormal events detected and addressed.
- **Workflows executed seamlessly:** Dependencies and parallel tasks managed without delays.

Path Cost:

The path cost indicates the cost of traveling from the starting state to the present state. In the AI Task Automation System, we focus on effective use of time and efficiency:

- **Time cost:** Total time required to complete all tasks.
- **Resource cost:** Energy, material, and labor consumption.
- **Error cost:** Penalties for human errors, machine failures, or workflow disruptions.
- **Priority weight:** Importance of completing high-priority tasks earlier.

The AI searches for the path (sequence of actions) that minimizes total path cost while ensuring all tasks are completed, machines are maintained, and resources are used efficiently.

SOLUTION

The solution defines the systematic route the AI Task Automation System follows to transition from an initial state to a goal state. Each action directly supports task movement through defined state progressions, ensuring accurate task execution and optimized performance.

1. **Assign high-priority tasks** (KR1 – Task Assignment and Prioritization)
    - AI evaluates task list, worker skills, and machine availability.
    - Assigns critical production tasks to qualified operators and machines.
    - Updates task status from "pending" → "in progress."

2. **Trigger dependent tasks** (KR4 – Workflow Automation Rules)
    - Once a prerequisite task is completed, AI automatically starts dependent tasks.
    - Example: After raw material processing (Task A) is complete, assembly task (Task B) begins.

3. **Monitor machine conditions** (KR2 – Machine Condition and Maintenance Monitoring)
    - AI continuously checks sensors for temperature, vibration, or other anomalies.
    - Predictive maintenance is scheduled if thresholds are exceeded.
    - Prevents unexpected machine downtime.

4. **Optimize resource allocation** (KR3 – Resource Utilization Optimization)
    - AI balances energy consumption, material usage, and manpower allocation.
    - Reassigns tasks to avoid overloading specific workers or machines.

5. **Detect anomalies and notify supervisors** (KR5 – Anomaly Detection and Reporting)
   - If a process behaves abnormally (e.g., delayed completion, machine irregularity), AI alerts the supervisor.
   - Corrective actions are taken immediately to prevent workflow disruption.

6. **Repeat for remaining tasks**
   - AI continuously cycles through the above actions, dynamically updating assignments, triggering workflows, and monitoring resources.
   - Parallel tasks are executed simultaneously where possible to reduce completion time.

CONCLUSION

The formulation of the AI Task Automation System as a state-space search problem demonstrates how intelligent decision-making can streamline industrial operations and overcome long-standing manual inefficiencies. By defining clear initial states, actions, goals, and path costs, the system is able to navigate through complex operational environments and choose the most efficient sequence of steps to reach the desired outcome.

Through the integration of Knowledge Representation (KR1–KR5), the AI gains structured reasoning capabilities that allow it to assign tasks automatically, synchronize workflows, predict machine failures, optimize resources, and detect anomalies in real time. The resulting sequence of actions—from assigning workers to triggering dependent processes—shows how the system transitions smoothly between states until all production goals are met.

Overall, this approach ensures that industrial operations become more efficient, accurate, and reliable. The state space formulation confirms that the AI system can continuously evaluate conditions, adapt to changes, and guide the entire workflow toward a fully optimized goal state. By combining logical KR rules with intelligent state transitions, the AI Task Automation System provides a robust foundation for smarter, safer, and more productive industrial environments.

PEER REVIEW

Peer Review on Team Work	
Reviewer: Tahiya Nuzhath Khan	
Team Members	**Category**
Tahiya Nuzhath Khan	4
Aria Fardhin Nidhi	4
Pashmia Binte Walid	4
Malika Kasmalieva	4